DILI — Acetaminophen (APAP) Toxicity

1. AST/ALT >10,000 furnace

WHAT YOU SEE

Blazing furnace reading AST >10,000, ALT >10,000

WHAT IT ENCODES

Acetaminophen produces massive centrilobular (zone 3) hepatocellular necrosis, so aminotransferases climb into the thousands and often exceed 10,000 IU/L — far higher than viral or alcoholic hepatitis. Only a handful of insults reach this 'four-figure' range: APAP, ischemic ('shock') liver, and acute viral hepatitis. ALT and AST are markers of injury, not function; the prognostic markers are INR/PT, bilirubin, lactate, pH and creatinine.

BOARD TRAP

WRONG to attribute transaminases >10,000 to alcoholic hepatitis. Alcoholic hepatitis classically keeps AST <300–500 with an AST:ALT ratio >2 (both rarely above ~8x ULN); sky-high transaminases instead point to APAP toxicity or ischemic hepatitis. The discriminator is the magnitude plus the AST:ALT ratio.

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2. 2 g alcoholic threshold scale

WHAT YOU SEE

Scale with 2 g and skull = ALCOHOLIC THRESHOLD

WHAT IT ENCODES

Chronic alcoholics, malnourished, and fasting patients develop toxicity at much lower doses (~2–4 g/day) because chronic alcohol induces CYP2E1 (shunting more APAP to toxic NAPQI) and poor intake depletes hepatic glutathione (the NAPQI buffer). The same applies to chronic isoniazid or anticonvulsant users. In these patients, even 'therapeutic' doses can be hepatotoxic.

BOARD TRAP

WRONG to assume a low/therapeutic APAP dose excludes toxicity in a malnourished alcoholic — induced CYP2E1 plus depleted glutathione lowers the toxic threshold to ~2–4 g/day. The discriminator is the host: glutathione stores and CYP2E1 status, not just the milligrams ingested.

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3. 10-15 g general population

WHAT YOU SEE

Scale arm marked 10–15 g GENERAL POPULATION

WHAT IT ENCODES

In a healthy adult, the toxic single-acute-ingestion threshold is roughly >150 mg/kg or >7.5–10 g; ingestions of 10–15 g reliably cause hepatotoxicity and >20–25 g can be fatal without treatment. Risk-stratify by dose AND by the 4-hour level on the Rumack-Matthew nomogram for a single timed ingestion.

BOARD TRAP

WRONG to use the dose alone to decide on NAC for a single acute ingestion when a timed level is obtainable — plot the 4-hour level on the nomogram. Conversely, for staggered/chronic ingestion the nomogram fails and you treat based on dose history and labs (>150 mg/kg or symptoms/elevated transaminases).

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4. 3-4 g/day SAFE bottle

WHAT YOU SEE

Man holding bottle: 3–4 g/day — SAFE?

WHAT IT ENCODES

The maximum therapeutic dose is up to ~4 g/day in healthy adults, commonly capped at 2–3 g/day in chronic liver disease, the elderly, malnutrition, or heavy alcohol use. A single tablet is typically 325–500 mg (extra-strength 500 mg). APAP is preferred over NSAIDs for analgesia in CKD and in those at GI/bleeding risk.

BOARD TRAP

WRONG to call 4 g/day universally 'safe' — in cirrhosis, malnutrition, or chronic alcohol use the safe ceiling drops to 2–3 g/day, and unrecognized combination products (opioid/APAP, OTC cold remedies) push patients over it. The discriminator is the host and hidden sources of APAP.

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5. 4H BEST star

WHAT YOU SEE

Gold star: 8H BEST / 4H draw

WHAT IT ENCODES

Draw the serum APAP level at 4 hours post-ingestion — the earliest reliable point (absorption complete) for plotting on the Rumack-Matthew nomogram. NAC is most effective when started within 8 hours of ingestion, so timing the level and starting NAC promptly is critical; if results will be delayed past 8 h, start NAC empirically and stop it if the level falls below the line.

BOARD TRAP

WRONG to plot a level drawn BEFORE 4 hours on the nomogram — pre-4-hour levels are uninterpretable because absorption is incomplete. The discriminator: never use early or untimed levels to withhold NAC; when in doubt about timing, treat.

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6. 24-36H STILL WORKS clock

WHAT YOU SEE

Green sign 24–36 H STILL WORKS, crossing out TOO LATE

WHAT IT ENCODES

N-acetylcysteine is most effective within 8 hours but still benefits patients who present late (>24 h) and even those with established hepatotoxicity or acute liver failure, where it improves transplant-free survival. Never withhold NAC for late presentation — give it whenever toxicity is suspected, and continue until INR and transaminases improve and the patient is clinically stable.

BOARD TRAP

WRONG to withhold NAC because 'it's too late' or because >24 hours have passed — NAC still improves outcomes in late presentation and established liver failure. The discriminator is suspected/confirmed APAP toxicity, not the clock; the only role of timing is that earlier is better.

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7. Viral hep / ischemic crossed out

WHAT YOU SEE

VIRAL HEP? and ISCHEMIC? faces with red X, hemodynamic event note

WHAT IT ENCODES

The differential for transaminases in the thousands is short: acetaminophen, ischemic ('shock') hepatitis after a hypotensive/hemodynamic event, and acute viral hepatitis. Ischemic hepatitis shows a rapid rise-and-fall of AST/ALT with a high LDH and the AST often peaking first after a documented hypotensive episode; viral hepatitis is supported by serologies and a more sustained course.

BOARD TRAP

WRONG to overlook ischemic hepatitis in a patient with transient hypotension/cardiac arrest and transaminases >5,000 — the rapid rise-and-fall pattern with markedly elevated LDH (low ALT:LDH ratio) distinguishes shock liver from viral hepatitis and points away from APAP when there's no ingestion history.

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8. NAC 140 mg/kg load plant

WHAT YOU SEE
Plant/flask labeled 140 mg/kg LOAD

WHAT IT ENCODES
N-acetylcysteine is the antidote because it replenishes hepatic glutathione (and directly scavenges NAPQI / acts as an antioxidant). Oral protocol: 140 mg/kg loading dose, then 70 mg/kg every 4 hours for 17 doses. The common IV protocol is a 3-bag regimen over ~21 hours; IV is preferred in vomiting, hepatic failure, or pregnancy.

BOARD TRAP
WRONG to delay NAC waiting for the nomogram level in a clearly high-risk or vomiting patient — start empirically and de-escalate later if the level is below the line. Anaphylactoid reactions to IV NAC are rate-related and managed by slowing the infusion, not by abandoning the antidote.

9. 70 mg/kg Q4H x17

WHAT YOU SEE
Plant labeled 70 mg/kg Q4H x17

WHAT IT ENCODES
The classic oral NAC maintenance regimen is 70 mg/kg every 4 hours for 17 doses after the 140 mg/kg load. Modern practice often uses weight-based IV NAC and extends therapy beyond the fixed course if the patient still has rising transaminases, coagulopathy, or detectable APAP — i.e., treat to clinical/biochemical recovery, not just to a dose count.

BOARD TRAP
WRONG to stop NAC strictly at 17 doses (or end of the 21-hour IV course) if the patient still has rising INR/transaminases or a detectable APAP level — continue NAC until liver tests are improving and APAP is undetectable. The discriminator is the trajectory of INR/transaminases, not the protocol counter.

10. 30 min max / renal exit

WHAT YOU SEE
Funnel: 30 MIN MAX, RENAL EXIT

WHAT IT ENCODES
Acetaminophen is rapidly absorbed with a peak at ~30–60 minutes (delayed by co-ingested anticholinergics/opioids or extended-release formulations). Most APAP is conjugated (glucuronidation/sulfation) and excreted renally; only a small fraction goes through CYP2E1 to the toxic NAPQI, which is normally neutralized by glutathione.

BOARD TRAP
WRONG to anchor on a single early/4-hour level after an extended-release product or a co-ingested opioid/anticholinergic — delayed or erratic absorption can push the peak later, so repeat levels are needed. The discriminator is the formulation and co-ingestants altering absorption kinetics.

11. 10 JAUNDICE figures

WHAT YOU SEE
Group of 10 yellow jaundiced figures on scale

WHAT IT ENCODES
Hy's Law: drug-induced hepatocellular injury (ALT >3x ULN) PLUS jaundice (total bilirubin >2x ULN) without biliary obstruction predicts a high mortality — roughly 1 in 10 such patients progress to acute liver failure or death. It is the key prognostic rule for serious DILI from any hepatotoxic drug, not just APAP.

BOARD TRAP
WRONG to think a mildly elevated ALT alone defines dangerous DILI — Hy's Law requires the combination of hepatocellular injury AND jaundice (with a normal alk phos / R-ratio favoring hepatocellular pattern). Isolated transaminitis without hyperbilirubinemia carries a far better prognosis; the bilirubin is the discriminator.

12. 1 ALF skull scale

WHAT YOU SEE
Lawyer holding scale: 1 ALF with skull

WHAT IT ENCODES
Acetaminophen is the #1 cause of acute liver failure in the US and the developed world. King's College criteria flag the need for transplant: arterial pH <7.3 (after resuscitation), OR the triad of INR >6.5 (PT >100 s) + creatinine >3.4 mg/dL + grade III/IV hepatic encephalopathy. A lactate >3.5 mmol/L is an additional poor-prognosis marker.

BOARD TRAP
WRONG to reassure based on improving transaminases in fulminant APAP failure — falling AST/ALT can simply mean there are few hepatocytes left to leak enzyme. The discriminator for outcome is synthetic/clearance function: a rising INR, worsening encephalopathy, acidosis, and creatinine (King's College criteria), not the transaminase number.

13. BLACK BOX WARNING table

WHAT YOU SEE
Document: BLACK BOX WARNING (crossed)

WHAT IT ENCODES
APAP's chief safety concern is dose-dependent hepatotoxicity, which drives its hepatic warnings and the limits on combination products. Hidden acetaminophen in opioid/APAP combinations (e.g., hydrocodone-APAP) and OTC cold/flu remedies is a major cause of accidental overdose, prompting limits on per-tablet APAP content.

BOARD TRAP
WRONG to overlook total acetaminophen from multiple products — a patient taking an opioid/APAP combination plus an OTC cold remedy can unknowingly exceed 4 g/day. The discriminator is summing every APAP-containing product, since accidental staggered overdose is treated by dose history and labs (the nomogram does not apply).

14. Stages timeline (clocks)

WHAT YOU SEE
Clock progression 0H -> 20H -> 24-36H across the top

WHAT IT ENCODES
Four clinical stages: I (0–24 h) nausea/vomiting/malaise, often asymptomatic with normal labs; II (24–72 h) RUQ pain with rising AST/ALT, bilirubin and INR; III (72–96 h) peak hepatotoxicity — maximal transaminases, jaundice, coagulopathy, encephalopathy, possible AKI; IV (4 days–2 weeks) recovery if the patient survives. NAC is most useful before stage III.

BOARD TRAP
WRONG to be reassured by a well-appearing patient with normal labs in stage I (0–24 h) — early-presenting overdose patients look deceptively well before injury manifests. The discriminator is the timeline plus a 4-hour APAP level/dose history; do not defer NAC just because the patient is currently asymptomatic with normal transaminases.